

Q. Prove that  $V(aX+bY) = a^2 V(X) + b^2 V(Y) + 2ab \text{Cov}(X, Y)$ .

$$\begin{aligned}\text{So! } V(aX+bY) &= E[(aX+bY)^2] - [E(aX+bY)]^2 \\&= E[a^2 X^2 + b^2 Y^2 + 2abXY] - [aE(X) + bE(Y)]^2 \\&= a^2 E(X^2) + b^2 E(Y^2) + 2ab E(XY) \\&\quad - [a^2 E(X)^2 + b^2 E(Y)^2 + 2ab E(X)E(Y)] \\&= a^2 [E(X^2) - E(X)^2] + b^2 [E(Y^2) - E(Y)^2] \\&\quad + 2ab [E(XY) - E(X)E(Y)] \\&= a^2 V(X) + b^2 V(Y) + 2ab \text{Cov}(X, Y)\end{aligned}$$

Q. Suppose that the two dimensional r.v  $(X, Y)$  has joint pdf

$$f(x, y) = kx(x-y), \quad 0 < x < 2, \quad -x < y < x,$$

a) Evaluate the constant  $k$ .

b) Find the marginal pdf of  $X$

c) Find the marginal pdf of  $Y$

$$\text{Sol: a) } \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$$

$$\Rightarrow \int_{x=0}^{x=2} \int_{y=-x}^{y=x} k(x^2 - xy) dy dx = 1$$

$$\Rightarrow k \int_0^2 \left[ x^2 y - \frac{xy^2}{2} \right]_{y=-x}^{y=x} dx = 1$$

$$\Rightarrow k \int_0^2 \left[ x^3 - \frac{x^3}{2} \right] - \left[ -x^3 - \frac{x^3}{2} \right] dx = 1$$

$$\Rightarrow k \int_0^2 x^3 - \frac{x^3}{2} + x^3 + \frac{x^3}{2} dx = 1$$

$$\Rightarrow 2k \left[ \frac{x^4}{4} \right]_0^2 = \frac{k}{2} \times 16 = 1$$

$$\Rightarrow k = \frac{1}{8}$$

b) We have  $f(x, y) = \frac{x(x-y)}{8}$   $\begin{matrix} 0 < x < 2 \\ -x < y < x \end{matrix}$

Marginal pdf of  $x$  is  $g(x) = \int_{-\infty}^{\infty} f(x, y) dy$ .

Since  $f(x, y) = 0$  if  $x \notin (0, 2)$ ,  
 $g(x) = 0$  if  $x \notin (0, 2)$ .

For  $0 < x < 2$ ,

$$g(x) = \int_{-x}^x \frac{x(x-y)}{8} dy = \frac{1}{8} \int_{-x}^x (x^2 - xy) dy$$

$$= \frac{1}{8} \left[ x^2 y - \frac{xy^2}{2} \right]_{-x}^x$$

$$= \frac{1}{8} \left[ \left( x^3 - \frac{x^3}{2} \right) - \left( -x^3 - \frac{x^3}{2} \right) \right] = \frac{2x^3}{8} = \frac{x^3}{4}$$

Marginal pdf of  $x$  is  $g(x) = \frac{x^3}{4}$   $0 < x < 2$

$= 0$  elsewhere.

c) Marginal pdf of  $Y$  is  $h(y) = \int_{-\infty}^{\infty} f(x, y) dx$

Since  $f(x, y) = 0$  when  $y \notin (-x, x)$ ,  
 $h(y) = 0$  when  $y \notin (-x, x)$ .

We need to find  $h(y)$  for  $-x < y < x$ .

Case i)

If  $-x < y < 0 \Rightarrow 0 < -y < x$ . But  $x < 2$   
 $\Rightarrow 0 < -y < x < 2$

$$\therefore h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_{-y}^2 \frac{x(x-y)}{8} dx$$

$$= \frac{1}{8} \times \left[ \frac{x^3}{3} - \frac{x^2 y}{2} \right]_{-y}^2$$

$$= \frac{1}{8} \left[ \left( \frac{8}{3} - \frac{4y}{2} \right) - \left( -\frac{y^3}{3} + \frac{y^3}{2} \right) \right]$$

$$= \frac{1}{3} - \frac{y}{4} + \frac{y^3}{24} - \frac{y^3}{16}$$

$$= \frac{1}{3} - \frac{y}{4} - \frac{y^3}{48}$$

Case ii)

I,  $0 \leq y < x$ . Since  $x < 2$ ,  $y < x < 2$

$$\begin{aligned}\therefore h(y) &= \int_y^2 \frac{1}{8} x(x-y) dx \\ &= \frac{1}{8} \left[ \frac{x^3}{3} - \frac{x^2 y}{2} \right]_y^2 = \frac{1}{8} \left[ \left( \frac{8}{3} - 2y \right) - \left( \frac{y^3}{3} - \frac{y^3}{2} \right) \right] \\ &= \frac{1}{3} - \frac{y}{4} - \frac{y^3}{24} + \frac{y^3}{16} = \frac{1}{3} - \frac{y}{4} + \frac{5}{48} y^3\end{aligned}$$

$$\therefore h(y) = \begin{cases} \frac{1}{3} - \frac{y}{4} - \frac{y^3}{48} & , \text{ if } -x < y < 0 \\ \frac{1}{3} - \frac{y}{4} + \frac{5}{48} y^3 & , \text{ if } 0 \leq y < x \\ 0 & \text{ otherwise.} \end{cases}$$

Q. Suppose that the joint pdf of two dimensional r.v.  $(X, Y)$  is given by

$$f(x, y) = x^2 + \frac{xy}{3}, \quad 0 < x < 1, \quad 0 < y < 2$$

$$= 0, \quad \text{elsewhere}$$

Compute following:

a)  $P(X > \frac{1}{2})$

b)  $P(Y < X)$

$$c) P(Y < \frac{1}{2} \mid X < \frac{1}{2}) \quad d) P(X+Y > 1)$$

$$\text{Sol: a) } P(X > \frac{1}{2}) = P(X > \frac{1}{2}, -\infty < Y < \infty)$$

$$\text{We know, } P[(X, Y) \in B] = \iint_B f(x, y) dx dy.$$

$$\therefore P(X > \frac{1}{2}) = \int_{-\infty}^{\infty} \int_{\frac{1}{2}}^{\infty} \left( x^2 + \frac{xy}{3} \right) dx dy$$

$$= \int_0^2 \int_{\frac{1}{2}}^1 \left( x^2 + \frac{xy}{3} \right) dx dy$$

$$= \int_0^2 \left[ \frac{x^3}{3} + \frac{x^2 y}{6} \right]_{\frac{1}{2}}^1 dy$$

$$= \int_0^2 \left[ \left( \frac{1}{3} + \frac{y}{6} \right) - \left( \frac{1}{24} + \frac{y}{24} \right) \right] dy$$

$$= \left[ \frac{y}{3} + \frac{y^2}{12} - \frac{y}{24} - \frac{y^2}{48} \right]_0^2$$

$$= \frac{2}{3} + \frac{4}{12} - \frac{2}{24} - \frac{4}{48} = \frac{2}{3} + \frac{1}{3} - \frac{1}{12} - \frac{1}{12}$$

$$= 1 - \frac{1}{6} = \frac{6-1}{6} = \frac{5}{6}$$

b) To find  $P(Y < X) = P(-\infty < X < \infty, Y < X)$

$$= \int_{x=-\infty}^{x=\infty} \int_{y=-\infty}^{y=x} f(x, y) dy dx = \int_0^1 \int_0^x \left( x^2 + \frac{xy}{3} \right) dy dx$$

$$= \int_0^1 \left[ x^2 y + \frac{xy^2}{6} \right]_{y=0}^{y=x} dx$$

$$= \int_0^1 \left[ x^3 + \frac{x^3}{6} \right] dx = \int_0^1 \frac{7x^3}{6} dx = \frac{7}{6} \times \frac{1}{4} = \frac{7}{24}$$

$$c) P(Y < \frac{1}{2} | X < \frac{1}{2}) = \frac{P(\{Y < \frac{1}{2}\} \cap \{X < \frac{1}{2}\})}{P(X < \frac{1}{2})}$$

$$= \frac{P(X < \frac{1}{2}, Y < \frac{1}{2})}{P(X < \frac{1}{2})}$$

$$P(X < \frac{1}{2}, Y < \frac{1}{2}) = \int_{y=-\infty}^{y=\frac{1}{2}} \int_{x=-\infty}^{x=\frac{1}{2}} \left( x^2 + \frac{xy}{3} \right) dx dy$$

$$= \int_{y=0}^{y=1/2} \int_{x=0}^{x=1/2} \left( x^2 + \frac{xy}{3} \right) dx dy$$

$$= \int_0^{1/2} \left[ \frac{x^3}{3} + \frac{x^2 y}{6} \right]_0^{1/2} dy = \int_0^{1/2} \left[ \frac{1}{24} + \frac{y}{24} \right] dy$$

$$= \left[ \frac{y}{24} + \frac{y^2}{48} \right]_0^{1/2} = \frac{1}{48} + \frac{1}{4 \times 48}$$

$$= \frac{1}{48} \times \left( 1 + \frac{1}{4} \right) = \frac{5}{192}$$

Now  $P(X < 1/2) = 1 - P(X > 1/2)$

$$= 1 - \frac{5}{6} = \frac{1}{6}$$

$$\therefore P(Y < 1/2 \mid X < 1/2) = \frac{5/192}{1/6} = \underline{\underline{\frac{5}{32}}}$$

d)  $g(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^2 \left( x^2 + \frac{xy}{3} \right) dy$



$$= \left[ x^2 y + \frac{xy^2}{6} \right]_0^2 = 2x^2 + \frac{4x}{6} = 2x^2 + \frac{2x}{3} = 2 \left( x^2 + \frac{x}{3} \right)$$

$$c) h(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^1 \left( x^2 + \frac{xy}{3} \right) dx$$

$$= \left[ \frac{x^3}{3} + \frac{x^2 y}{6} \right]_0^1 = \frac{1}{3} + \frac{y}{6}$$

$$f) P(X+Y > 1)$$

$$\text{Given } X+Y > 1 \Leftrightarrow Y > 1-X.$$

$$\begin{aligned} \therefore P(X+Y > 1) &= P(Y > 1-X) \\ &= P(Y > 1-X, -\infty < X < \infty) \end{aligned}$$

$$= \int_{x=-\infty}^{x=\infty} \int_{y=1-x}^{y=\infty} \left( x^2 + \frac{xy}{3} \right) dy dx$$

$$= \int_0^1 \int_{y=1-x}^{y=2} \left( x^2 + \frac{xy}{3} \right) dy dx$$

$$= \int_0^1 \left[ x^2 y + \frac{xy^2}{6} \right]_{y=1-x}^{y=2} dx$$

$$= \int_0^1 \left[ 2x^2 + \frac{4x}{6} \right] - \left[ x^2(1-x) + \frac{x(1-x)^2}{6} \right] dx$$

$$= \int_0^1 \left[ 2x^2 + \frac{2x}{3} - x^2 + x^3 - \frac{x(1+x^2-2x)}{6} \right] dx$$

$$= \int_0^1 \left[ 2x^2 + \frac{2x}{3} - x^2 + x^3 - \frac{x}{6} - \frac{x^3}{6} + \frac{2x^2}{6} \right] dx$$

$$= \left[ \frac{2x^3}{3} + \frac{2x^2}{6} - \frac{x^3}{3} + \frac{x^4}{4} - \frac{x^2}{12} - \frac{x^4}{24} + \frac{x^3}{9} \right]_0^1$$

$$= \frac{2}{3} + \frac{2}{6} - \frac{1}{3} + \frac{1}{4} - \frac{1}{12} - \frac{1}{24} + \frac{1}{9}$$

$$= \frac{16}{24} + \frac{6}{24} - \frac{2}{24} - \frac{1}{24} + \frac{1}{9}$$

$$= \frac{16+6-3}{24} + \frac{1}{9} = \frac{19}{24} + \frac{1}{9} = \frac{195}{216} = 0.9028$$

Q. Using the joint pmf table of discrete r.v  $(X, Y)$ , Determine the  $\text{cov}(X, Y)$

|         | $X = 1$        | $X = 2$        | $X = 3$        |
|---------|----------------|----------------|----------------|
| $Y = 1$ | $\frac{1}{12}$ | $\frac{1}{12}$ | $\frac{1}{12}$ |
| $Y = 2$ | $\frac{1}{4}$  | $\frac{1}{4}$  | $\frac{1}{4}$  |

Sol.: The marginal pmf of  $X$  &  $Y$  are given by following table

|         | $X = 1$                | $X = 2$                | $X = 3$                | $P_Y$                  |
|---------|------------------------|------------------------|------------------------|------------------------|
| $Y = 1$ | $\frac{1}{12}$         | $\frac{1}{12}$         | $\frac{1}{12}$         | $P_Y(1) = \frac{1}{4}$ |
| $Y = 2$ | $\frac{1}{4}$          | $\frac{1}{4}$          | $\frac{1}{4}$          | $P_Y(2) = \frac{3}{4}$ |
| $P_X$   | $P_X(1) = \frac{1}{3}$ | $P_X(2) = \frac{1}{3}$ | $P_X(3) = \frac{1}{3}$ |                        |

$$E(X) = \sum x p_x = 1 \times \frac{1}{3} + 2 \times \frac{1}{3} + 3 \times \frac{1}{3} = \frac{6}{3} = 2$$

$$E(Y) = \sum y p_y = 1 \times \frac{1}{4} + 2 \times \frac{3}{4} = \frac{7}{4}$$

Expectation of  $E(XY) = \sum \sum xy p_{xy}(x, y)$

$$= 1 \times 1 \times \frac{1}{12} + 2 \times 1 \times \frac{1}{12} + 3 \times 1 \times \frac{1}{12}$$

$$+ 1 \times 2 \times \frac{1}{4} + 2 \times 2 \times \frac{1}{4} + 3 \times 2 \times \frac{1}{4}$$

$$= \frac{1+2+3}{12} + \frac{2+4+6}{4} = \frac{1}{2} + 3 = \frac{7}{2}$$

$$\begin{aligned} \therefore \text{Cov}(X, Y) &= E(XY) - E(X)E(Y) \\ &= \frac{7}{2} - 2 \times \frac{7}{4} = \underline{\underline{0}} \end{aligned}$$

Q. Suppose the following table represents the joint pmf of discrete r.v.  $(X, Y)$ . Evaluate the marginal pmf.

| $Y \backslash X$ | 1              | 2             | 3              |
|------------------|----------------|---------------|----------------|
| 1                | $\frac{1}{12}$ | $\frac{1}{6}$ | 0              |
| 2                | 0              | $\frac{1}{9}$ | $\frac{1}{5}$  |
| 3                | $\frac{1}{18}$ | $\frac{1}{4}$ | $\frac{2}{15}$ |

Sol:

Marginal pmf of  $X$  is  $p_X(x) = \sum_{i=1}^3 p(x, y_i)$

Marginal pmf of  $Y$  is  $p_Y(y) = \sum_{i=1}^3 p(x_i, y)$

| $Y \backslash X$ | 1                       | 2                        | 3                        | $p_Y$                     |
|------------------|-------------------------|--------------------------|--------------------------|---------------------------|
| 1                | $\frac{1}{12}$          | $\frac{1}{6}$            | 0                        | $p_Y(1) = \frac{1}{4}$    |
| 2                | 0                       | $\frac{1}{9}$            | $\frac{1}{5}$            | $p_Y(2) = \frac{14}{45}$  |
| 3                | $\frac{1}{18}$          | $\frac{1}{4}$            | $\frac{2}{15}$           | $p_Y(3) = \frac{79}{180}$ |
| $p_X$            | $p_X(1) = \frac{5}{36}$ | $p_X(2) = \frac{19}{36}$ | $p_X(3) = \frac{12}{36}$ |                           |